DEMONSTRATION ONLY — NOT A QUALIFIED PROCEDURE; THE FACILITY OWNER'S QUALIFIED PROCEDURES AND APPLICABLE CODES GOVERN.

WO-100-04 — Remake flanged joint 2"-P300-LS-3007/A2 on 2"-P300-LS-3007

Area	A100 Tank farm	**Task**	GJ Flanged joint remake
Priority	1 (safety)	**Issued by**	Facility owner (demonstration)
Primary asset	**T-101** — T-101 tank 12' dia × 22' high — E 25'-0" N 26'-6" EL 100'-6"		
Equipment	T-101	**Lines**	2"-P300-LS-3007
Isometrics	ISO-2"-P300-LS-3007 (generated); owner's drawings ISO-LS-3007		
Permits	General work permit		

Reason

Steam leak at the tank heating-coil inlet flange.

Scope of work

1. Isolate per the isolation list, drain and vent 2"-P300-LS-3007.
2. Break the joint at E 22'-0" N -12" EL 123'-9-1/4" between FLANGE WELD-NECK 2" P300 RAISED FACE and VALVE, GATE, WEDGE, OS&Y 2" P300.
3. Inspect both flange faces (SW-NDE-5).
4. Fit a new gasket and new studs; tighten to the stated preload in three passes, cross pattern (SW-WO-5).
5. Remove blinds; leak test on start-up (SW-TEST-2).

Affected items: 2"-P300-LS-3007/A2, 2"-P300-LS-3007/A1, 2"-P300-LS-3007/A3

Rules: SW-ITP-1, SW-ITP-2, SW-ITP-3, SW-NDE-5, SW-TEST-2, SW-WO-1, SW-WO-2, SW-WO-3, SW-WO-5, SW-WO-7

Owner's isometric drawings (SW-IMP-6)

Drawing	Line drawn	Work line	Sheets	Match to the plant model
ISO-LS-3007	2"-P300-LS-3007	2"-P300-LS-3007	1	agrees with the model

ISO-LS-3007 (PDF) — the generated isometrics follow the ITP.

Assets in the plant model (SW-WO-7)

Tag	Kind	Description	Line	Location
T-101	equipment	T-101 tank 12' dia × 22' high		E 25'-0" N 26'-6" EL 100'-6"
2"-P300-LS-3007/A2	item	GASKET 2" P300 COMPRESSED SHEET	2"-P300-LS-3007	E 22'-0" N -12" EL 123'-9-1/4"
2"-P300-LS-3007/A1	item	FLANGE WELD-NECK 2" P300 RAISED FACE	2"-P300-LS-3007	E 22'-0" N -12" EL 123'-7-1/8"
2"-P300-LS-3007/A3	item	VALVE, GATE, WEDGE, OS&Y 2" P300	2"-P300-LS-3007	E 22'-0" N -12" EL 124'-2-11/16"
6"-P300-LS-2003/T:2"-P300-LS-3007	item	TEE REDUCING 6" × 2" P300	6"-P300-LS-2003	E 22'-0" N -12" EL 123'-0"
2"-P300-LS-3007/B2	item	VALVE, GATE, WEDGE, OS&Y 2" P300	2"-P300-LS-3007	E 18'-5/8" N 26'-6" EL 102'-6"
2"-P300-LS-3007/B5	item	FLANGE WELD-NECK 2" P300 RAISED FACE	2"-P300-LS-3007	E 17'-5-1/16" N 26'-6" EL 102'-6"

Isolation list (SW-WO-3)

#	Kind	Tag	Line	Action	Location	Fitted ✓	Removed ✓
1	equipment	6"-P300-LS-2003/T:2"-P300-LS-3007	6"-P300-LS-2003	No valve on this side within the line: isolate or depressure the parent line 6"-P300-LS-2003 at its tee under the shutdown isolation			
2	valve	2"-P300-LS-3007/B2	2"-P300-LS-3007	Close and lock VALVE, GATE, WEDGE, OS&Y 2" P300 (2"-P300-LS-3007/B2)	E 18'-5/8" N 26'-6" EL 102'-6"		
3	blind	2"-P300-LS-3007/B5	2"-P300-LS-3007	Fit a line blind between 2"-P300-LS-3007/B2 and FLANGE WELD-NECK 2" P300 RAISED FACE (2"-P300-LS-3007/B5), work side	E 17'-7-1/8" N 26'-6" EL 102'-6"		

Parts

Description	Qty	Replaces
GASKET 2" P300 COMPRESSED SHEET	1	2"-P300-LS-3007/A2
STUD SET 4 × 0.625" × 4-1/4" WITH 2 NUTS, FOR 2" P300	1	2"-P300-LS-3007/A4

Inspections

Id	Inspection	Criteria	Basis
WO-100-04-I1	Flange faces before the new gasket goes in	No radial scratch across the sealing face (SW-NDE-5)	SW-NDE-5

Bolt-up (SW-WO-5)

Joint	Studs	Preload per stud (lbf)	Passes	Done ✓
2"-P300-LS-3007/A2	4 × 0.625" studs (2"-P300-LS-3007/A4)	13,806	3	

Test — Service leak test

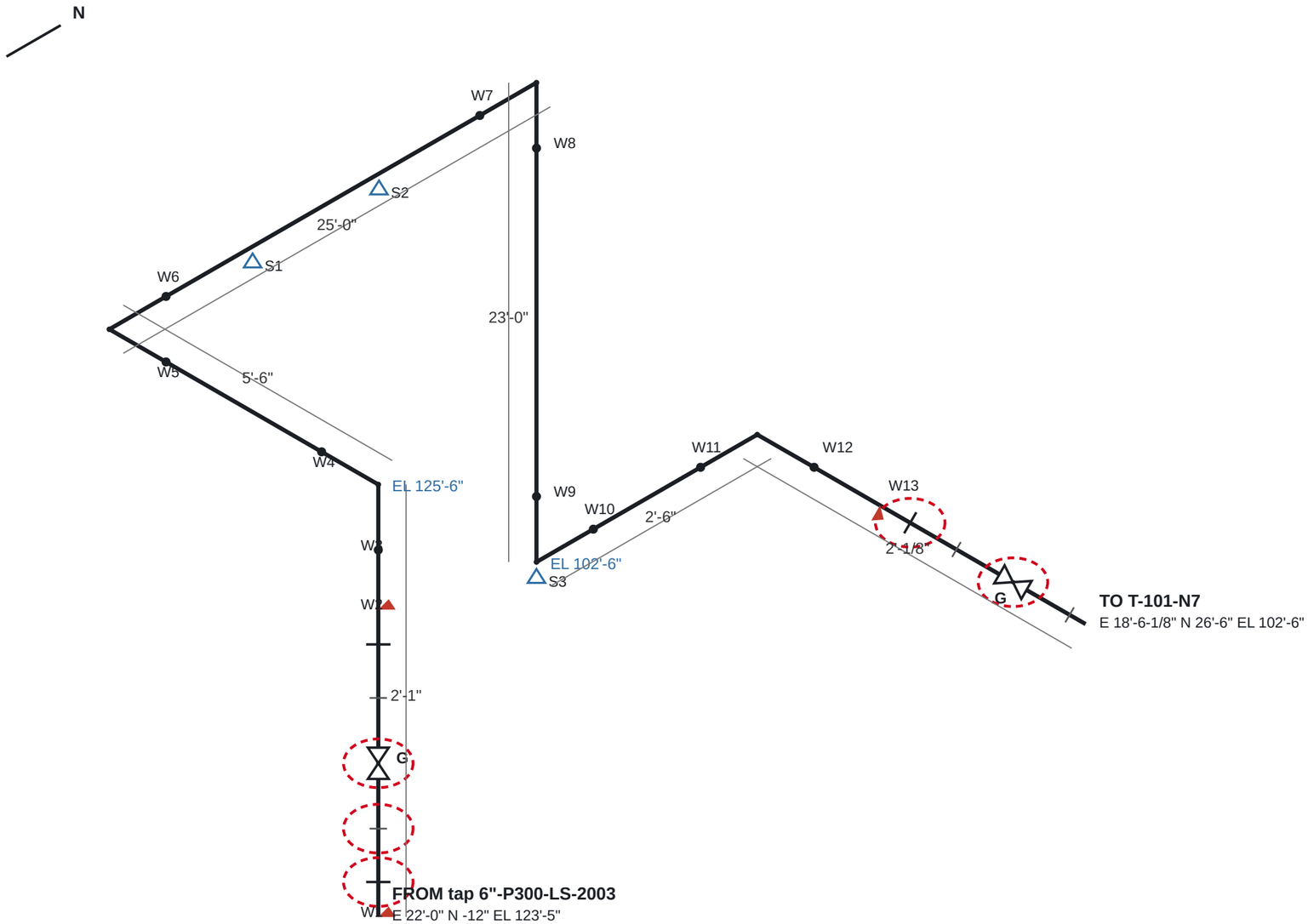
Description	The remade joint checked at operating pressure on start-up
Basis	SW-TEST-2

Inspection and test plan (SW-ITP-1..3)

H hold · W witness · R review · S surveillance. Status: accepted / rejected / not applicable.

#	Activity	Reference	Acceptance	Record	Contr.	Owner	3rd party	Status	Signed (contr. / owner / 3rd)	Date	Remark
1	Work permit issued; isolation list (valves locked, blinds fitted) verified at zero pressure and drained	SW-WO-3, isolation list	Every isolation in place and tagged; zero pressure at the vent	Signed isolation certificate	H	H	—				
2	Joint broken; both flange faces inspected	SW-NDE-5	No radial scratch across the sealing face	Flange face inspection record	W	W	—				
3	New gasket fitted; studs tightened to preload in three passes	SW-WO-5	Preload per stud as stated	Bolt-up record	W	W	—				
4	Service leak test on start-up	SW-TEST-2	No leak	Leak test record	W	W	—				
5	Blinds removed, isolations returned to normal line-up per the isolation list	SW-WO-3	Every blind removed and accounted for	Isolation certificate closed	H	H	—				
6	Records complete; work order handed back to operations	SW-ITP-3	All ITP activities accepted or not applicable	Signed ITP and completion record	H	H	R				

ISOMETRIC — 2"-P300-LS-3007 — Remake flanged joint 2"-P300-LS-3007/A2 on 2"-P300-LS-3007



BILL OF MATERIALS

1	FLANGE WELD-NECK 2" P300 RAISED FACE	3
2	GASKET 2" P300 COMPRESSED SHEET	4
3	VALVE, GATE, WEDGE, OS&Y 2" P300	2
4	STUD SET 4 × 0.625" × 4-1/4" WITH 2 NUTS, FOR 2" P300	4
5	PIPE 2" P300 BEVELLED ENDS	54'-5-1/4"
6	ELBOW 90° R = 5/3 OD 2" P300	5

WELDS 13 (3 FIELD) · SUPPORTS 3 · SPOOLS 4

W1F W2F W3 W4 W5 W6 W7 W8 W9 W10
W11 W12 W13F

2"-P300-LS-3007

CLASS S300 — Low-pressure steam — INSULATED
SERVICE LS · 2" · P300 · DESIGN 300 psi
tap 6"-P300-LS-2003 → T-101-N7
PIPE 54'-5-1/4" · WEIGHT 297 lb

DWG ISO-2"-P300-LS-3007 REV 0 WO-100-04

SmartWeld AI · not to scale · E lower right, N upper right, EL up (SW-ISO-1)

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